Envelope Equations

USPAS - Beam Physics with Intense Space Charge

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Envelope equations: reduced-order characterization of beam dynamics in terms of second-order moments

[Image]

Road map

- Start from single particle equation of motion: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
- Make use of:
  - Paraxial (near-axis) approximation
  - Conservation of angular momentum
  - Axisymmetry of external forces
- Arrive at single-particle paraxial ray equation.
- Take statistical averages to obtain moment/envelope equations.

Paraxial ray equation

Equation of motion in cylindrical coordinates

Equation of motion for particle of mass m and charge q in electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$:

$$\frac{d\mathbf{p}}{dt} = \frac{d(\gamma m \dot{\mathbf{x}})}{dt} = q\left(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}\right), \tag{1}$$

Switch to cylindrical coordinates (r, θ, z):

$$\begin{aligned} \mathbf{p} &= p_x \hat{x} + p_y \hat{y} + p_z \hat{z} \\ &= p_r \hat{r} + p_\theta^* \hat{\theta} + p_z \hat{z}. \end{aligned} \tag{2}$$

The components p_r and p_θ^* are given by

$$\begin{aligned} p_r &= \gamma m \dot{r}, \\ p_\theta^* &= \gamma m r \dot{\theta}. \end{aligned} \tag{3}$$

The symbol p_θ^* is used to reserve p_θ for the *canonical* angular momentum, defined later.

In cylindrical coordinates, the unit vectors depend on the particle position and thus change with time:

$$
\begin{aligned}
\dot{\hat{r}} &\equiv d\hat{r}/dt = \dot{\theta}\hat{\theta}, \\
\dot{\hat{\theta}} &\equiv d\hat{\theta}/dt = -\dot{\theta}\hat{r}.
\end{aligned} \tag{4}
$$

Thus, the time derivative of the momentum vector in Eq. (2) is:

$$
\begin{aligned}
\frac{d\mathbf{p}}{dt} &= \frac{d}{dt}\left(p_r\hat{r} + p_\theta^*\hat{\theta} + p_z\hat{z}\right) && (5)\\
&= \dot{p}_r\hat{r} + p_r\dot{\hat{r}} + \dot{p}_\theta\hat{\theta} + p_\theta^*\dot{\hat{\theta}} + \dot{p}_z\hat{z} && (6)\\
&= \left(\dot{p}_r - p_\theta^*\dot{\theta}\right)\hat{r} + \left(\dot{p}_\theta + p_r\dot{\theta}\right)\hat{\theta} + \dot{p}_z\hat{z} && (7)\\
&= \left[\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2\right]\hat{r} + \left[\frac{d(\gamma m r\dot{\theta})}{dt} + \gamma m\dot{r}\dot{\theta}\right]\hat{\theta} + \left[\frac{d(\gamma m\dot{z})}{dt}\right]\hat{z}. && (8)\\
&= \left[\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2\right]\hat{r} + \left[\frac{1}{r}\frac{d}{dt}\left(\gamma m r^2\dot{\theta}\right)\right]\hat{\theta} + \left[\frac{d(\gamma m\dot{z})}{dt}\right]\hat{z}. && (9)
\end{aligned}
$$

This takes care of the left side of the Lorentz force equation (Eq. (1)).

The right side of Eq. (1) requires the cylindrical field components:

$$\begin{aligned} \mathbf{E} &= E_r\hat{r} + E_\theta\hat{\theta} + E_z\hat{z}, \\ \mathbf{B} &= B_r\hat{r} + B_\theta\hat{\theta} + B_z\hat{z}. \end{aligned} \tag{10}$$

Evaluating the cross product $\dot{\mathbf{x}} \times \mathbf{B}$,

$$\dot{\mathbf{x}} \times \mathbf{B} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ \dot{r} & r\dot{\theta} & \dot{z} \\ B_r & B_\theta & B_z \end{vmatrix}, \tag{11}$$

leads to

$$\frac{d(\gamma m\dot{r})}{dt} - \gamma m r\dot{\theta}^2 = q\left(E_r + r\dot{\theta}B_z - \dot{z}B_\theta\right), \tag{12}$$

$$\frac{1}{r}\frac{d(\gamma m r^2\dot{\theta})}{dt} = q\left(E_\theta + \dot{z}B_r - \dot{r}B_z\right), \tag{13}$$

$$\frac{d(\gamma m\dot{z})}{dt} = q\left(E_z + \dot{r}B_\theta - r\dot{\theta}B_r\right) \tag{14}$$

From fields to potentials

Write the fields in terms of the scalar potential $\phi(\mathbf{x})$ and vector potential $\mathbf{A}(\mathbf{x})$:

$$\begin{aligned} \mathbf{E} &= -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \\ \mathbf{B} &= \nabla \times \mathbf{A}. \end{aligned} \tag{15}$$

The gradient $\nabla\phi$ in cylindrical coordinates is:

$$\nabla\phi = \left[\frac{\partial\phi}{\partial r}\right]\hat{r} + \left[\frac{1}{r}\frac{\partial\phi}{\partial\theta}\right]\hat{\theta} + \left[\frac{\partial\phi}{\partial z}\right]\hat{z}. \tag{16}$$

And the curl $\nabla \times \mathbf{A}$ is:

$$\nabla \times \mathbf{A} = \left[\frac{1}{r}\frac{\partial A_z}{\partial\theta} - \frac{\partial A_\theta}{\partial z}\right]\hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right]\hat{z}. \tag{17}$$

Axial symmetry means all partial derivatives with respect to θ are zero in Eq. (16) and Eq. (17). This gives the following expressions for the fields in terms of the potentials in cylindrical coordinates:

$$\begin{aligned}
\mathbf{E} &= \left[-\frac{\partial \phi}{\partial r} - \frac{\partial A_r}{\partial t}\right]\hat{r} + \left[-\frac{\partial A_\theta}{\partial t}\right]\hat{\theta} + \left[-\frac{\partial \phi}{\partial z} - \frac{\partial A_z}{\partial t}\right]\hat{z}, \\
\mathbf{B} &= \left[-\frac{\partial A_\theta}{\partial r}\right]\hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\hat{\theta} + \left[\frac{1}{r}\frac{\partial (rA_\theta)}{\partial r}\right]\hat{z}.
\end{aligned} \tag{18}$$

Conservation of angular momentum

Use the potentials to rewrite the angular equation of motion (Eq. (13)):

$$
\begin{aligned}
\frac{d}{dt}\left(\gamma m r^2 \dot{\theta}\right) &= qr\left(E_\theta + \dot{z}B_r - \dot{r}B_z\right) \\
&= qr\left(\left[-\frac{\partial A_\theta}{\partial t}\right] + \dot{z}\left[-\frac{\partial A_\theta}{\partial z}\right] - \dot{r}\left[\frac{1}{r}\frac{\partial (rA_\theta)}{\partial r}\right]\right) \\
&= -q\left(\frac{\partial (rA_\theta)}{\partial t} + \dot{z}\frac{\partial (rA_\theta)}{\partial z} + \dot{r}\frac{\partial (rA_\theta)}{\partial r}\right) \\
&= -q\left[\frac{\partial}{\partial t} + \dot{z}\frac{\partial}{\partial z} + \dot{r}\frac{\partial}{\partial r}\right](rA_\theta) \\
&= -q\left[\frac{\partial}{\partial t} + \dot{\mathbf{x}}\cdot\nabla\right](rA_\theta) \\
&= -\frac{d(qrA_\theta)}{dt}.
\end{aligned} \tag{19}
$$

Eq. (19) can then be written as:

$$\frac{d}{dt}\left(\gamma m r^2 \dot{\theta} + q r A_\theta\right) = 0. \tag{20}$$

This motivates the definition of the *canonical angular momentum*:

$$p_\theta \equiv \gamma m r^2 \dot{\theta} + q r A_\theta, \tag{21}$$

such that $dp_\theta/dt = 0$. We may also write p_θ in terms of the magnetic flux $\psi(r)$ though a circle of radius r *[Figure]*:

$$\psi(r) = \int \mathbf{B} \cdot d\Omega = \int \nabla \times \mathbf{A} \cdot d\Omega = \oint \mathbf{A} \cdot d\mathbf{l} = 2\pi r A_\theta. \tag{22}$$

Therefore, Eq. (21) can be written

$$p_\theta \equiv \gamma m r^2 \dot{\theta} + \frac{q\psi(r)}{2\pi}. \tag{23}$$

Series expansion of axially symmetric fields

Consider external fields $\mathbf{E}$ and $\mathbf{B}$ with azimuthal symmetry ($\partial/\partial\theta = 0$). In the absence of charge and current densities,

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= 0, \\ \nabla \times \mathbf{B} &= 0. \end{aligned} \tag{24}$$

The vanishing curl means we can express the fields as

$$\begin{aligned} \mathbf{E} &= -\nabla\phi, \\ \mathbf{B} &= -\nabla\phi_m, \end{aligned} \tag{25}$$

for scalar potentials ϕ and ϕ_m. Additionally, the vanishing divergence leads to the Laplace equation for the scalar potential:

$$\nabla^2\phi = 0. \tag{26}$$

In cylindrical coordinates, the Laplace equation for the potential is:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right)+\frac{\partial^2 \phi}{\partial z^2}=0. \tag{27}$$

The azimuthal symmetry will lead to a simplified form of the scalar potential in terms of its on-axis derivatives along the z axis. First, expand the potential $\phi(r, z)$ as a power series,

$$\phi(r, z)=\sum_{n=0}^{\infty} h_{2n}(z) r^{2n}, \tag{28}$$

where the $h_{2n}(z)$ are functions of z. The on-axis potential is given by $h_0(z)=\phi(0, z)$.

The rotational symmetry eliminates all odd powers of r in the series.

At the origin ($r = 0$), we must have $E_r = B_r = 0$. If we allowed the $n = 1$ term $h_1(z)r$, we would have $E_r(0, z) = -\partial\phi(r, z)/\partial r|_{r=0} = h_1(z) \neq 0$.

The derivatives in the Laplace equation are:

$$\begin{aligned} \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) &= \sum_{n=1}^{\infty}(2n)^2 h_{2n}(z) r^{2n-2}, \\ \frac{\partial^2 \phi}{\partial z^2} &= \sum_{n=0}^{\infty} h''_{2n}(z) r^{2n}, \end{aligned} \tag{29}$$

where $h'(z) = \partial h(z)/\partial z$.

Thus, $\nabla^2\phi = 0$ becomes:

$$
\begin{aligned}
\sum_{n=1}^{\infty}(2n)^2 h_{2n}(z) r^{2n-2} + \sum_{n=0}^{\infty} h''_{2n}(z) r^{2n} = 0 \\
\sum_{n=0}^{\infty}(2n+2)^2 h_{2n+2}(z) r^{2n} + \sum_{n=0}^{\infty} h''_{2n}(z) r^{2n} = 0 \\
\sum_{n=0}^{\infty}\left[(2n+2)^2 h_{2n+2}(z) + h''_{2n}(z)\right] = 0
\end{aligned}
\tag{30}
$$

The result is a recurrence relation,

$$
h_{2n+2}(z) = -\frac{h''_{2n}(z)}{(2n+2)^2}, \tag{31}
$$

with which we can write all terms as derivatives of $h_0(z)$ with respect to z:

$$
\begin{aligned}
h_2(z) &= -\frac{1}{4}\frac{\partial^2 h_0(z)}{\partial z^2}, \\
h_4(z) &= +\frac{1}{64}\frac{\partial^4 h_0(z)}{\partial z^4}, \\
&\vdots
\end{aligned}
\tag{32}
$$

We may thus write the potential at all points in space using only the z-derivatives of the on-axis potential $\phi(0, z)$:

$$\phi(r, z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n}\phi(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n} . \tag{33}$$

We now express the fields in terms of the expanded scalar potentials. Write the magnetic field as $\mathbf{B} = B_r \hat{r} + B_z \hat{z}$. Using the series expansion of $\phi_m(r, z)$, we find for the z component of the field:

$$
\begin{aligned}
B_z(r, z) &= -\frac{\partial \phi_m}{\partial z} \\
&= -\frac{\partial}{\partial z}\left[\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} \phi_m(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^2\right] \\
&= -\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n+1} \phi_m(0, z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^2 .
\end{aligned} \tag{34}
$$

Define $B(z)$ as the on-axis value of $B_z(z)$:

$$
B(z) \equiv B_z(0, z) = -\frac{\partial \phi_m(0, z)}{\partial z}. \tag{35}
$$

Inserting Eq. (35) into Eq. (34) gives:

$$
\begin{aligned}
B_z(r,z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n}}{\partial z^{2n}} \left[-\frac{\partial \phi_m(0,z)}{\partial z} \right] \left(\frac{r}{2}\right)^2 \\
B_z(r,z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} B(z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^2 .
\end{aligned}
\tag{36}
$$

For the radial component, we find:

$$
B_r(r,z) = \sum_{n=1}^{\infty} \frac{(-1)^{n}}{n!(n-1)!} \frac{\partial^{2n-1} B(z)}{\partial z^{2n-1}} \left(\frac{r}{2}\right)^{2n-1} . \tag{37}
$$

A similar procedure gives the radial and longitudinal components of the electric field. Define $V(z)$ as the on-axis value of the electric potential ϕ:

$$V(z) \equiv \phi(0, z). \tag{38}$$

This gives:

$$\begin{aligned} E_z(r, z) &= -\frac{\partial \phi}{\partial z} = -\sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n+1} V(z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^{2n}. \\ E_r(r, z) &= -\frac{\partial \phi}{\partial r} = -\sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n-1)!} \frac{\partial^{2n} V(z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n-1}. \end{aligned} \tag{39}$$

Derivation of paraxial ray equation

We will now take the *paraxial* limit, assuming small transverse position and velocity: $\dot{r} \ll \dot{z}$, $r\dot{\theta} \ll \dot{z}$. If $v = (\dot{r}^2 + r^2\dot{\theta}^2) + \dot{z}^2)^{1/2}$ is the total velocity, the paraxial limit gives $\dot{z} = (v^2 - \dot{r}^2 - r^2\dot{\theta}^2)^{1/2} \approx v$.

To first order in r and r', the external fields are:

$$\begin{aligned}
E_r^{(ext)} &\approx \frac{1}{2}\frac{\partial^2 V(z)}{\partial z^2} r, \\
B_z^{(ext)} &\approx B(z), \\
B_\theta^{(ext)} &= 0 \quad (\text{since } \partial\phi_m/\partial\theta = 0).
\end{aligned} \tag{40}$$

Plug these truncated fields into the radial equation of motion (Eq. (12)):

$$\begin{aligned}
\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 &= q\left(E_r + r\dot{\theta}B_z - \dot{z}B_\theta\right), \\
&\approx q\left(\frac{1}{2}\frac{\partial^2 V(z)}{\partial z^2} r + r\dot{\theta}B(z)\right)
\end{aligned} \tag{41}$$

Use z as the independent variable: $dz = \beta c dt$. Using $' \equiv d/dz$ and dropping the $\approx$ symbol gives:

$$\beta c \frac{d\left(\gamma m \beta c r'\right)}{dz} - \gamma m r (\beta c \theta')^2 = q\left(\frac{1}{2}V''(z) r + r \beta c \theta' B(z)\right). \tag{42}$$

Expand the derivative in the first term and divide by $\gamma \beta^2 m c^2$:

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' - r\theta'^2 = \frac{q}{\gamma\beta^2 m c^2}\left(\frac{1}{2}V''(z) + \beta c \theta' B(z)\right) r. \tag{43}$$

Now use the definition of the canonical angular momentum p_θ (Eq. (21)) to eliminate θ', noting that the flux $\psi(r) = \pi r^2 B$:

$$\begin{aligned}\theta' &= \frac{\dot{\theta}}{\beta c} \\ &= \frac{1}{\beta c}\frac{1}{\gamma m r^2}\left(p_\theta - \frac{q\psi(r)}{2\pi}\right), \\ &= \frac{p_\theta}{\gamma\beta m c r^2} - \frac{\omega_c}{2\gamma\beta c}.\end{aligned} \tag{44}$$

We have defined the *cyclotron frequency* $\omega_c \equiv \frac{qB}{m}$.

Substitute Eq. (44) into Eq. (43):

$$r'' + \frac{(\gamma\beta)'}{\gamma\beta}r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma\beta m c}\right)^2 \frac{1}{r^3} = \frac{q}{\gamma m \beta^2 c^2}\frac{1}{2}V''(z)r \tag{45}$$

The rate of change in the particle energy is given by

$$\frac{d(\gamma mc^2)}{dt} = q\mathbf{E}\cdot\mathbf{v}. \tag{46}$$

Rearrange and switch from t to z. In the paraxial limit:

$$\begin{aligned} \gamma' &= \frac{q}{mc^2}\frac{\mathbf{E}\cdot\mathbf{v}}{v_z} && (47) \\ &= \frac{q}{mc^2}\frac{E_x v_x + E_y v_y + E_z v_z}{v_z} && (48) \\ &\approx \frac{q}{mc^2}E_z. && (49) \end{aligned}$$

Giving the approximate relationship:

$$\gamma'' \approx \frac{q}{mc^2}E_z' = -\frac{q}{mc^2}V''. \tag{50}$$

This approximation of λ'' leads to the *paraxial ray equation*:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2}\right) r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma\beta mc}\right)^2 \frac{1}{r^3} = 0.} \qquad (51)$$

Terms in order: (i) inertial term; (ii) accelerative damping; (iii) E_r from converging field lines; (iv) solenoidal focusing ($v_\theta B_z$, part of centrifugal term); (v) centrifugal defocusing.

Addition of self-generated fields

We can also account for linearized self-fields $\mathbf{E}^{(self)}$ and $\mathbf{B}^{(self)}$. Adding these self-generated fields to the radial equation of motion gives:

$$r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} - r\theta'^2 = \frac{q}{mc^2\gamma\beta^2}\left(\frac{1}{2}V''(z)r + \beta c\theta' B(z)r + E_r^{(self)} - \beta c B_\theta^{(self)}\right). \tag{52}$$

So we only need the r component of the self-generated electric field and the θ component of the self-generated magnetic field. We will ignore the $B_z^{(self)}$ term in the paraxial limit.

Recall that $V(z) \equiv \phi(0, z)$ is the external potential ϕ on-axis, and to linear order $E_r(r, z) \approx V''(z)r/2$. Similarly, $B(z) \equiv B_z(0, z)$ is the external magnetic field on-axis, and to linear order $B_z(r, z) \approx B(z)$.

Assuming cylindrical symmetry of the beam, the Poisson equation for the self-generated electric potential is:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi^{(self)}}{\partial r}\right)+\frac{\partial^2 \phi^{(self)}}{\partial s^2}=-\frac{\rho(r,z)}{\epsilon_0}. \tag{53}$$

Solving for $\partial \phi^{(self)}/\partial r$ relates the radial electric field to z-derivative of the potential:

$$E_r^{(self)}=-\frac{\partial \phi^{(self)}}{\partial r}=\frac{\lambda(r,z)}{2\pi\epsilon_0 r}+\frac{1}{2}\frac{\partial^2 \phi^{(self)}}{\partial z^2}r, \tag{54}$$

where λ is the line-charge density,

$$\lambda(r,z)\equiv\int_0^r 2\pi\rho(\tilde{r},z)\tilde{r}d\tilde{r}. \tag{55}$$

Above, it is assumed that $\partial^2\phi^{(self)}/\partial z^2$ has no dependence on r (keep lowest-order term).

For the θ component of the self-generated magnetic field, use Ampere's law:

$$\oint \mathbf{B}^{(self)} \cdot d\mathbf{l} = \mu_0 I_{enc} \tag{56}$$

$$2\pi r B_\theta^{(self)} = \mu_0 \int_0^r 2\pi \tilde{r} J_z(\tilde{r}, z) d\tilde{r} = \mu_0 \lambda(r, z) v_z \tag{57}$$

$$B_\theta^{(self)} = \frac{\mu_0 \beta c \lambda(r, z)}{2\pi r} = \frac{\beta}{c} \frac{\lambda(r, z)}{2\pi \epsilon_0 r}. \tag{58}$$

We can now evaluate $E_r^{(self)} - \beta c B_\theta^{(self)}$:

$$E_r^{(self)} - \beta c B_\theta^{(self)} = \frac{1}{\gamma^2} \frac{\lambda(r)}{2\pi\epsilon_0 r} + \frac{1}{2} \frac{\partial^2 \phi^{(self)}}{\partial z^2} r, \tag{59}$$

We can also repeat the approximation of γ'' with the inclusion of the self-generated electric field.

$$\gamma'' \approx -\frac{q}{mc^2} \frac{\partial^2}{\partial z^2} \left(V + \phi^{(self)}\right). \tag{60}$$

The result is a driving term on the right side of the paraxial ray equation:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2}\right) r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma\beta mc}\right)^2 \frac{1}{r^3} = \frac{q}{mc^2\beta^2\gamma^3} \frac{\lambda(r)}{2\pi\epsilon_0 r}} \tag{61}$$

Envelope equations for axially symmetric beams

Time-evolution of expected value

Let $g(\mathbf{x}, \mathbf{x}')$ be any function of the phase space coordinates. Use brackets to indicate the expected value of g under probability distribution f:

$$\langle g \rangle \equiv \iint g(\mathbf{x}, \mathbf{x}') f(\mathbf{x}, \mathbf{x}', s) d\mathbf{x} d\mathbf{x}'. \tag{62}$$

What is the time-evolution of $\langle g \rangle$?

$$\frac{d\langle g \rangle}{ds} = \frac{d}{ds} \iint g(\mathbf{x}, \mathbf{x}') f(\mathbf{x}, \mathbf{x}', s) d\mathbf{x} d\mathbf{x}' \tag{63}$$

$$
\begin{aligned}
\frac{d\langle g\rangle}{ds} &= \frac{d}{ds}\iint g(\mathbf{x},\mathbf{x}')f(\mathbf{x},\mathbf{x}')d\mathbf{x}d\mathbf{x}' \\
&= \iint g(\mathbf{x},\mathbf{x}')\frac{\partial f(\mathbf{x},\mathbf{x}',s)}{\partial s}d\mathbf{x}d\mathbf{x}' \quad \textit{(Leibniz Rule)} \\
&= \iint g(\mathbf{x},\mathbf{x}')\left[-\mathbf{x}'\cdot\frac{\partial f}{\partial \mathbf{x}} - \mathbf{x}''\cdot\frac{\partial f}{\partial \mathbf{x}'}\right]d\mathbf{x}d\mathbf{x}' \quad \textit{(Vlasov equation)}.
\end{aligned}
\tag{64}
$$

Since f vanishes at the integration boundary (∞), we can use integration by parts to flip the gradients from f to g:

$$\begin{aligned}\iint \left[\mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} g\right] d\mathbf{x} d\mathbf{x}' &= -\iint \left[\mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} f\right] d\mathbf{x} d\mathbf{x}' \\ &= -\langle \mathbf{x}' \cdot \partial g / \partial \mathbf{x} \rangle.\end{aligned} \tag{65}$$

$$\begin{aligned}\iint \left[\mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} g\right] d\mathbf{x} d\mathbf{x}' &= -\iint \left[\mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'} f\right] d\mathbf{x} d\mathbf{x}' \\ &= -\langle \mathbf{x}'' \cdot \partial g / \partial \mathbf{x}' \rangle.\end{aligned} \tag{66}$$

Integration by parts (one-dimensional):

$$\int_a^b f(x)g'(x)dx = [f(x)g(x)]_a^b - \int_a^b f'(x)g(x)dx. \tag{67}$$

$$
\begin{aligned}
. &= \iint g(\mathbf{x}, \mathbf{x}')\mathbf{x}' \cdot \frac{\partial f(\mathbf{x}, \mathbf{x}', s)}{\partial \mathbf{x}} d\mathbf{x} d\mathbf{x}' \\
&= \iiiint g(x, x', y, y') \left[x' \frac{\partial}{\partial x} + y' \frac{\partial}{\partial y} \right] f(x, x', y, y', s) dx dx' dy dy' \\
&= \iiint x' \left[\int g \frac{\partial f}{\partial x} dx \right] dx' dy dy' + \iiint y' \left[\int g \frac{\partial f}{\partial y} dy \right] dx dx' dy' \\
&= -\iiiint f \left[x' \frac{\partial g}{\partial x} + x' \frac{\partial g}{\partial x} \right] dx dx' dy dy' \\
&= -\iint f(\mathbf{x}, \mathbf{x}', s) \left[\mathbf{x}' \cdot \frac{\partial g(\mathbf{x}, \mathbf{x}')}{\partial \mathbf{x}} \right] d\mathbf{x} d\mathbf{x}' \\
&= -\langle \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} \rangle.
\end{aligned}
\tag{68}
$$

We can then write $d\langle g \rangle / ds$ as:

$$\frac{d\langle g \rangle}{ds} = \langle \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} \rangle + \langle \mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'} \rangle. \tag{69}$$

From the definition of the total derivative:

$$\frac{dg}{ds} = \mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} + \mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'}. \tag{70}$$

Therefore, the time-derivative of the expected value is the same as the expected value of the time-derivative:

$$\frac{d\langle g \rangle}{ds} = \langle \frac{dg}{ds} \rangle. \tag{71}$$

To evaluate $d\langle g\rangle/ds$, we can evaluate dg/ds and then perform the averaging. For example:

$$\frac{d}{ds}\langle xx\rangle = 2\langle xx'\rangle \tag{72}$$

$$\frac{d}{ds}\langle x'x'\rangle = 2\langle x'x''\rangle \tag{73}$$

$$\frac{d}{ds}\langle xx'\rangle = \langle xx''\rangle + \langle x'x'\rangle \tag{74}$$

$$\vdots \tag{75}$$

Evolution of second moments

Define root-mean-square (rms) radius $\tilde{r}$:

$$\tilde{r} \equiv \sqrt{\langle r^2 \rangle}. \tag{76}$$

Evaluate $\tilde{r}'$:

$$\begin{aligned}
\tilde{r}^2 &= \langle r^2 \rangle \\
\frac{d}{ds}\tilde{r}^2 &= \frac{d}{ds}\langle r^2 \rangle \\
2\tilde{r}\tilde{r}' &= 2\langle r r' \rangle \\
\tilde{r}' &= \frac{\langle r r' \rangle}{\tilde{r}}.
\end{aligned} \tag{77}$$

Evaluate $\tilde{r}''$:

$$\begin{aligned}
\tilde{r}'' &= \frac{\langle rr'' \rangle + \langle r'r' \rangle}{\tilde{r}} - \frac{\langle rr' \rangle \tilde{r}'}{\tilde{r}^2} \\
&= \frac{\langle rr'' \rangle}{\tilde{r}} + \frac{\langle rr \rangle \langle r'r' \rangle - \langle rr' \rangle \langle rr' \rangle}{\tilde{r}^3} \\
&= \frac{\langle rr'' \rangle}{\tilde{r}} + \frac{\varepsilon_r}{\tilde{r}^3},
\end{aligned} \tag{78}$$

where we have defined the *rms radial emittance* ε_r:

$$\varepsilon_r \equiv \langle rr \rangle \langle r'r' \rangle - \langle rr' \rangle \langle rr' \rangle. \tag{79}$$

What is $\langle rr'' \rangle$?

Evaluation of $\langle rr'' \rangle$

Return to the paraxial ray equation for the radius r (including linear self-fields):

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2}\right) r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma\beta m c}\right)^2 \frac{1}{r^3} = \frac{q}{mc^2\beta^2\gamma^3} \frac{\lambda(r)}{2\pi\epsilon_0 r}} \tag{80}$$

Inserting Eq. (**??**) Eq. (**??**) gives the following *rms envelope equation*: [...]

$$\boxed{r_b'' + \left(\frac{\gamma'}{\gamma\beta^2}\right) r_b' + \left(\frac{\gamma''}{2\gamma\beta^2}\right) r_b + \left(\frac{\omega_c}{2\gamma\beta c}\right) r_b - \left(\frac{2\langle p_\theta\rangle}{\gamma\beta mc}\right)^2 \frac{1}{r_b^3} - \frac{\varepsilon_r^2}{r_b^3} - \frac{Q}{r_b},} \tag{81}$$

where Q is the *perveance*:

$$Q \equiv [\dots], \tag{82}$$

and ε_r is the *radial emittance*:

$$\begin{aligned} \varepsilon_r^2 &\equiv 4\left(\langle r^2\rangle\langle r'^2\rangle - \langle rr'\rangle^2 + \langle r^2\rangle\langle r^2\theta'^2\rangle - \langle r^2\theta'\rangle^2\right) \\ \varepsilon_r^2 &= \varepsilon_x^2 - 4\langle r^2\theta'\rangle^2. \end{aligned} \tag{83}$$

Envelope equations for elliptically symmetric beams

References